Medicine Outcome Prediction System

Deep model evaluation and comparison | Reproducible test-set analysis

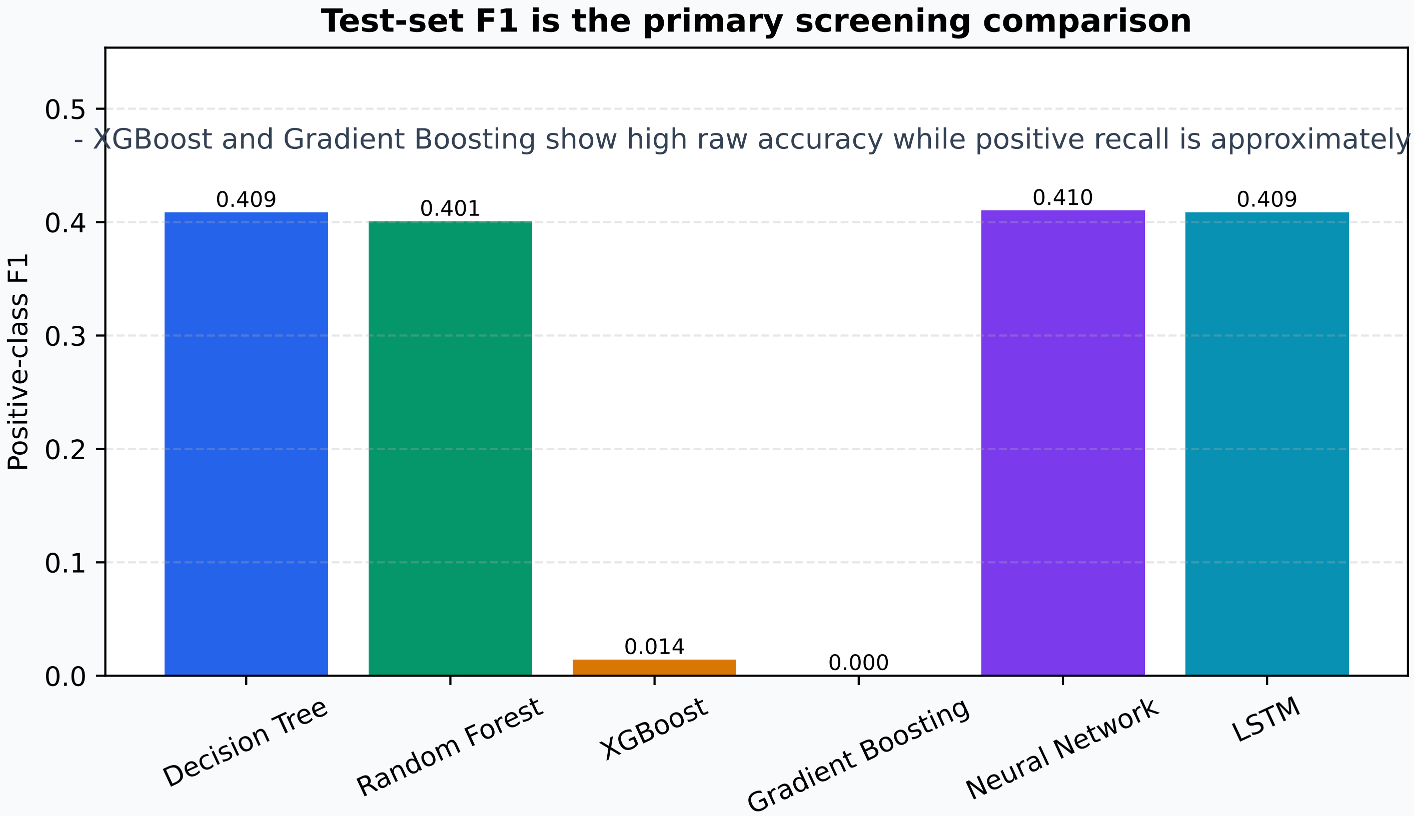
EXECUTIVE FINDINGS

- Best positive-class F1: Neural Network (0.410).

- Best PR-AUC: Neural Network (0.333); this is more informative than raw accuracy for the imbalanced o

- Best positive recall: Neural Network (0.641); recall matters when missing a Yes outcome is costly.

- Best ROC-AUC: Neural Network (0.646), but ranking ability does not guarantee a useful 0.5 threshold.



- XGBoost and Gradient Boosting show high raw accuracy while positive recall is approximately zero: the

Decision warning

These are predictive results, not evidence of clinical efficacy or treatment causality. Human review, calibration, subgroup testing, and prospective validation are required before use.

Test-Set Scorecard

All models evaluated on the identical 197,176-record test partition

Model	Acc	Bal acc	Prec	Recall	Spec	F1	ROC-AUC	PR-AUC	Brier
Decision Tree	0.591	0.606	0.302	0.632	0.579	0.409	0.642	0.326	0.235
Random Forest	0.589	0.598	0.297	0.615	0.582	0.401	0.633	0.316	0.238
XGBoost	0.777	0.503	0.508	0.007	0.998	0.014	0.646	0.331	0.166
Gradient Boosting	0.777	0.500	0.286	0.000	1.000	0.000	0.644	0.331	0.166
Neural Network	0.588	0.607	0.302	0.641	0.573	0.410	0.646	0.333	0.233
LSTM	0.599	0.607	0.305	0.620	0.594	0.409	0.646	0.331	0.232

Interpretation

Accuracy is not sufficient here: the positive class is the minority outcome. Balanced accuracy, positive recall, F1, PR-AUC, and the confusion matrix should drive selection. The Brier score measures probability error; lower is better. Confidence intervals below are bootstrap 95% intervals.

Model	Accuracy 95% CI	Balanced accuracy 95% CI
Decision Tree	0.5890 - 0.5931	0.6027 - 0.6079
Random Forest	0.5869 - 0.5910	0.5956 - 0.6006
XGBoost	0.7743 - 0.7784	0.5022 - 0.5031
Gradient Boosting	0.7743 - 0.7783	0.5000 - 0.5000
Neural Network	0.5863 - 0.5902	0.6046 - 0.6093
LSTM	0.5973 - 0.6013	0.6042 - 0.6090